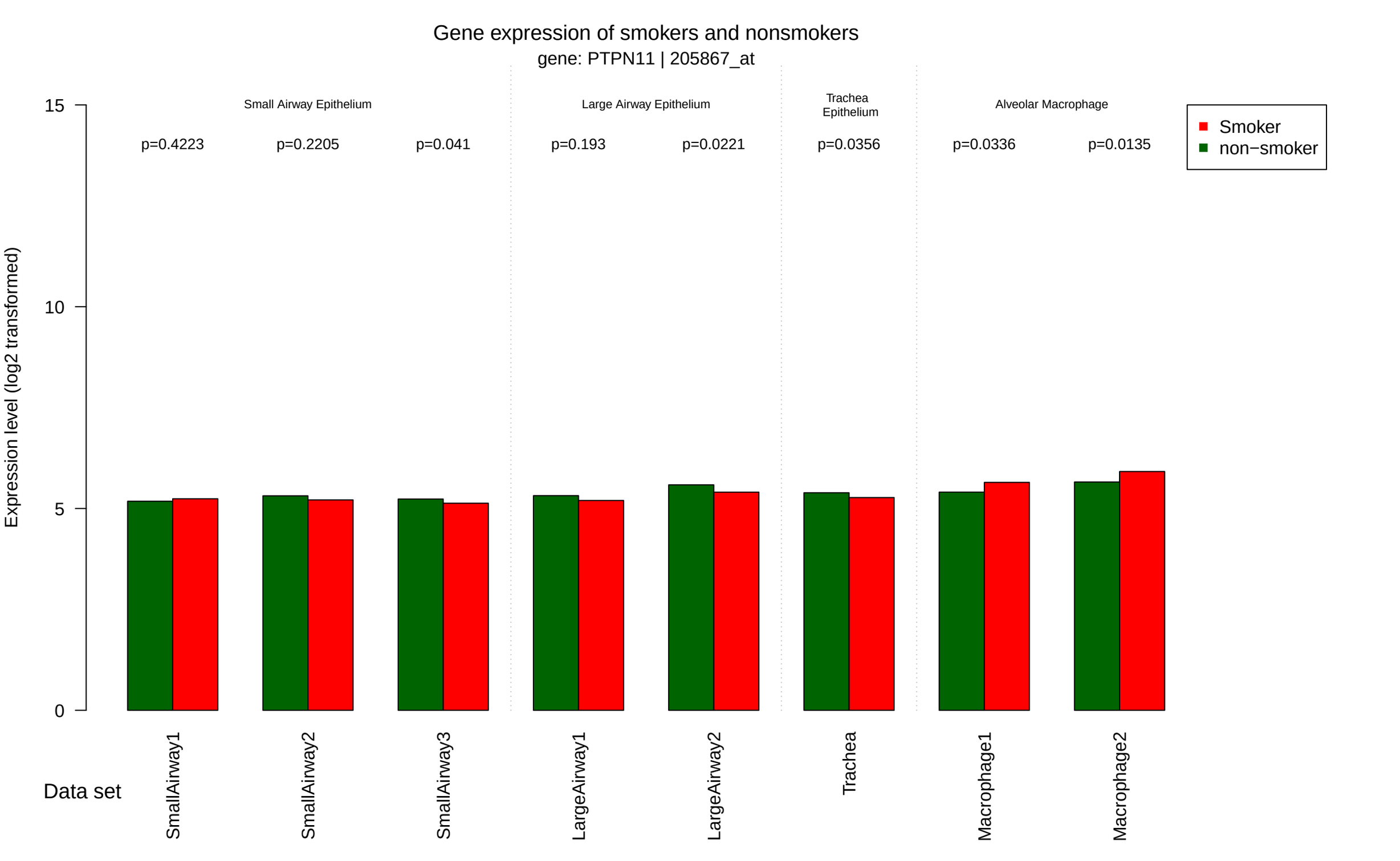
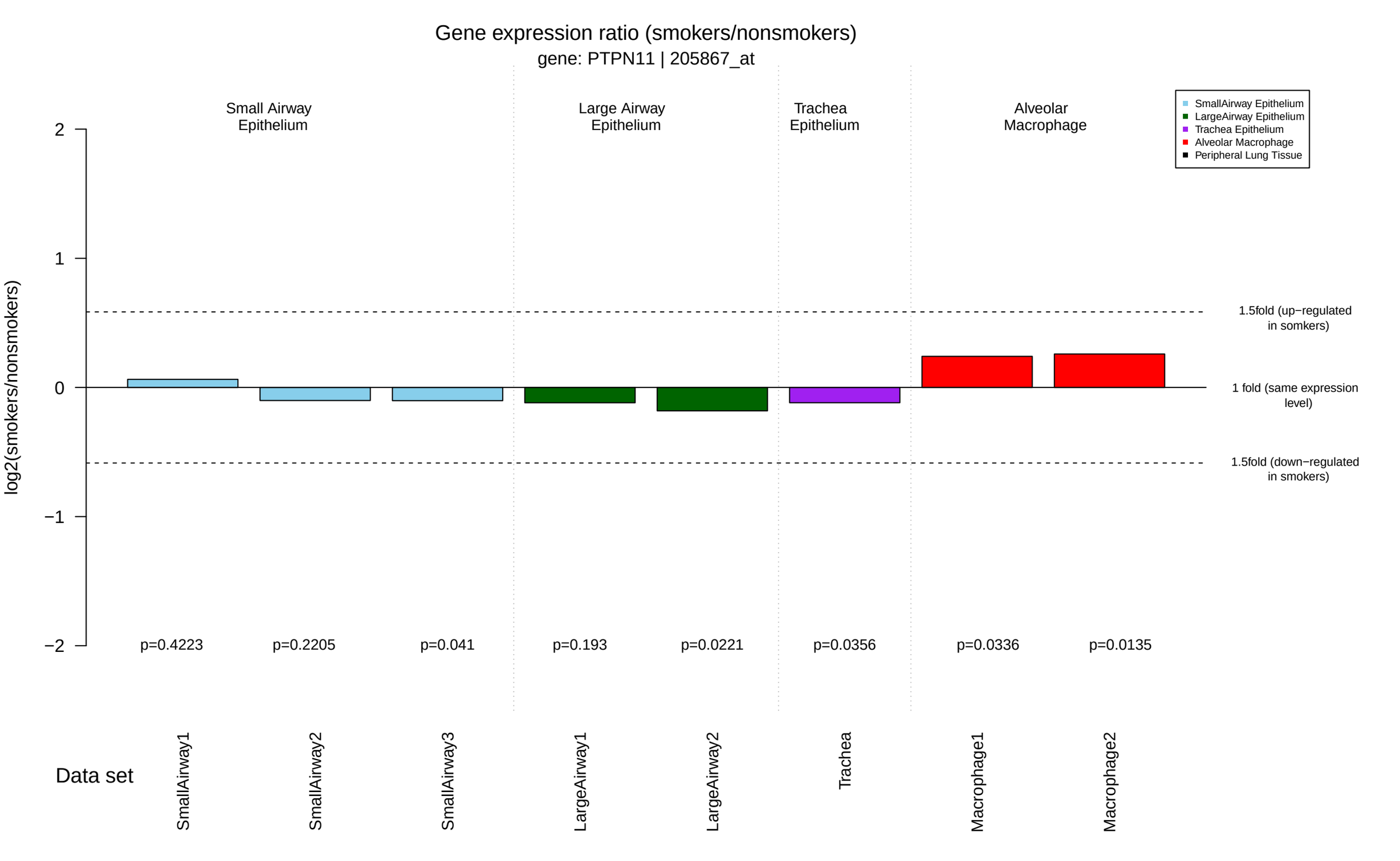
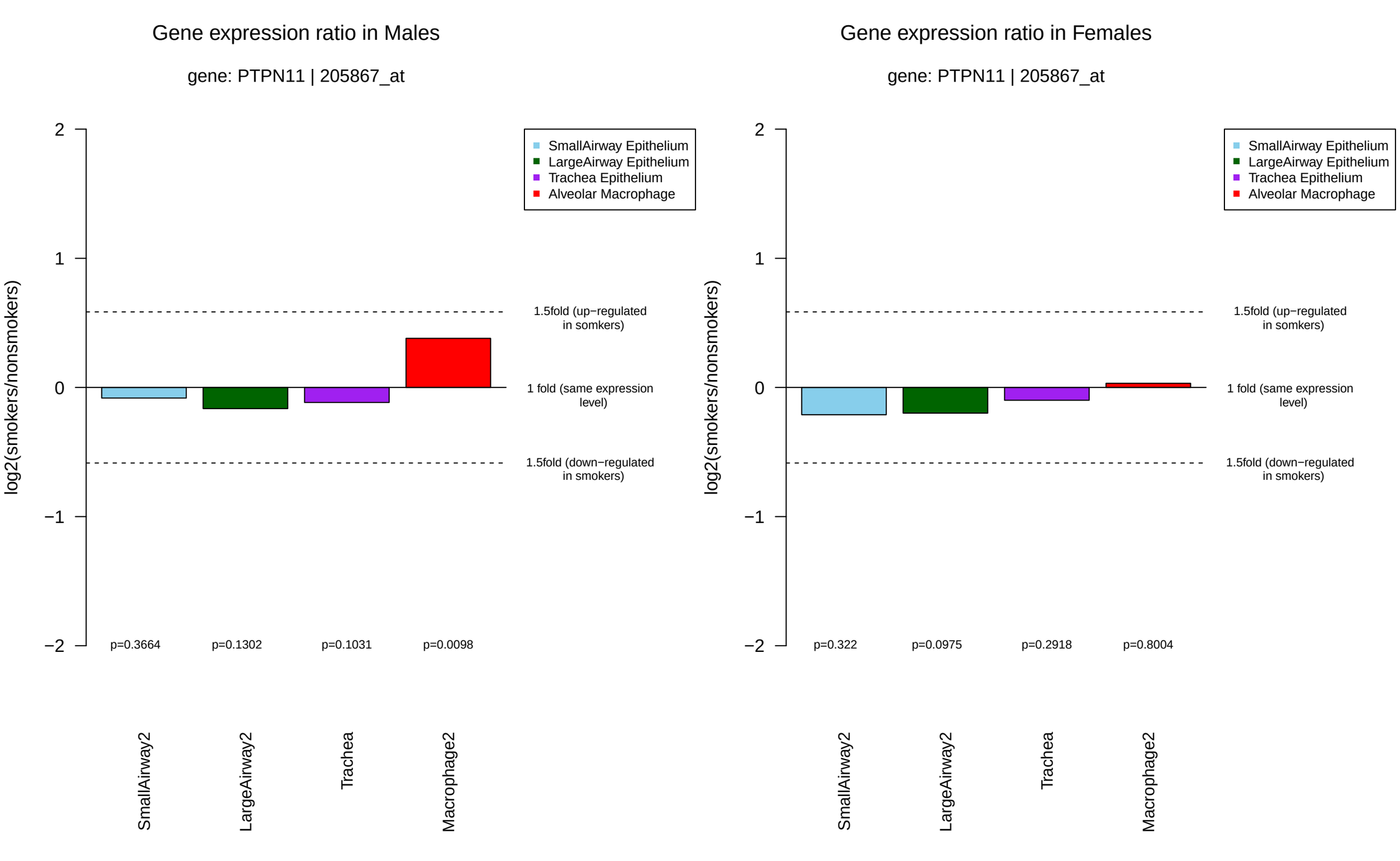
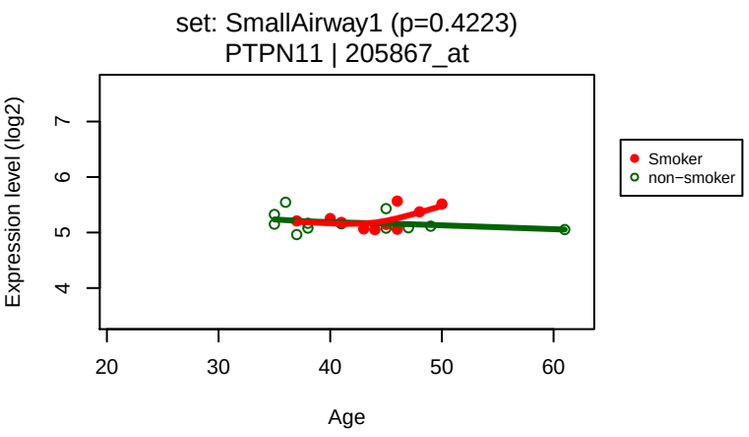
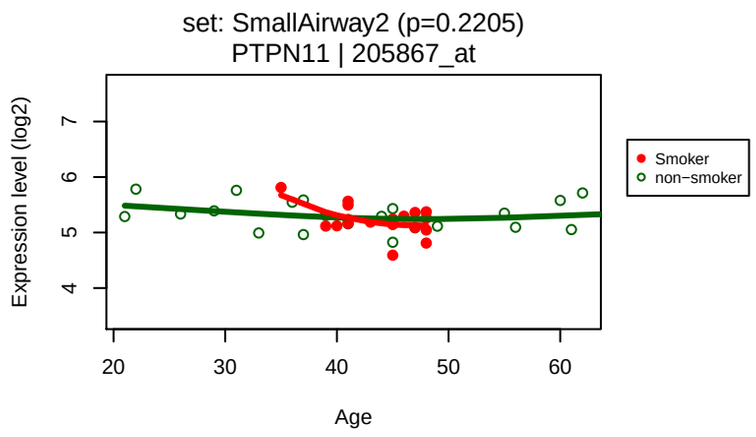
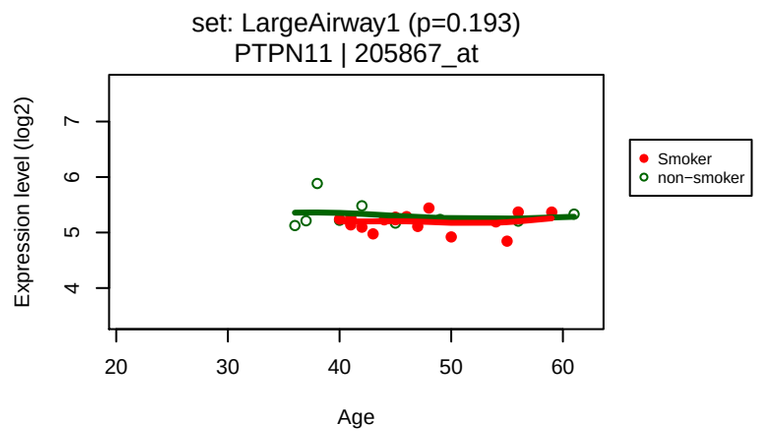

Gene expression of smokers and nonsmokers
gene: PTPN11 | 205867_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4223
p=0.2205
p=0.041
p=0.193
p=0.0221
p=0.0356
p=0.0336
p=0.0135
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

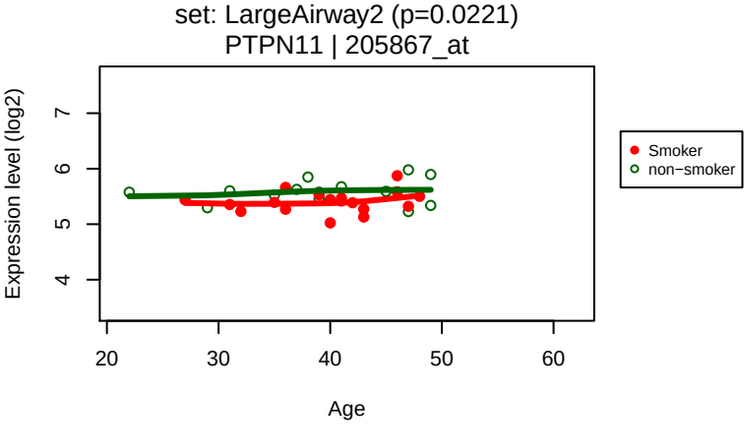

Gene expression ratio (smokers/nonsmokers)
gene: PTPN11 | 205867_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4223
p=0.2205
p=0.041
p=0.193
p=0.0221
p=0.0356
p=0.0336
p=0.0135
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

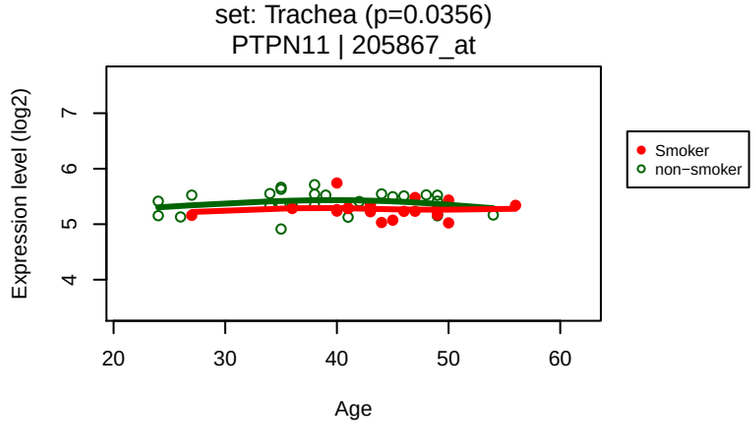

Gene expression ratio in Males
gene: PTPN11 | 205867_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3664
p=0.1302
p=0.1031
p=0.0098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN11 | 205867_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.322
p=0.0975
p=0.2918
p=0.8004
SmallAirway2
LargeAirway2
Trachea
Macrophage2

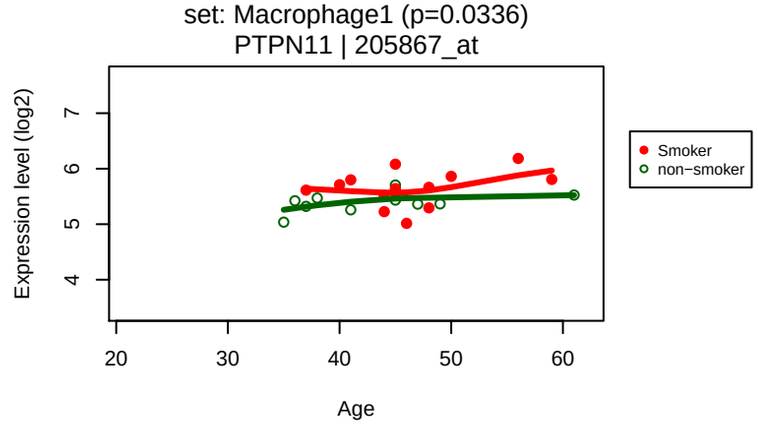

set: SmallAirway1 (p=0.4223)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker

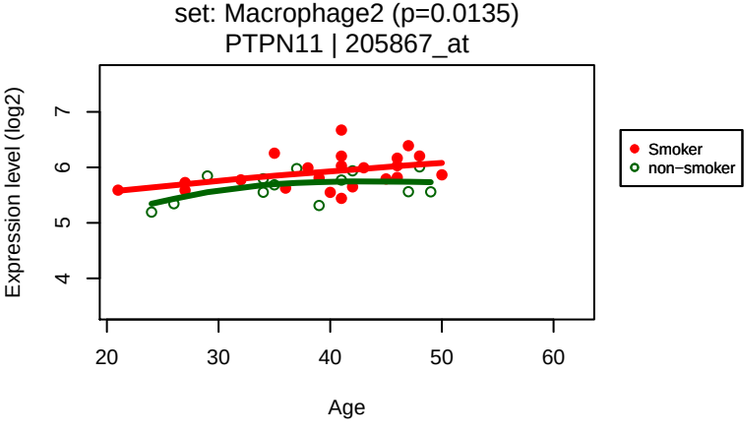

set: SmallAirway2 (p=0.2205)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker

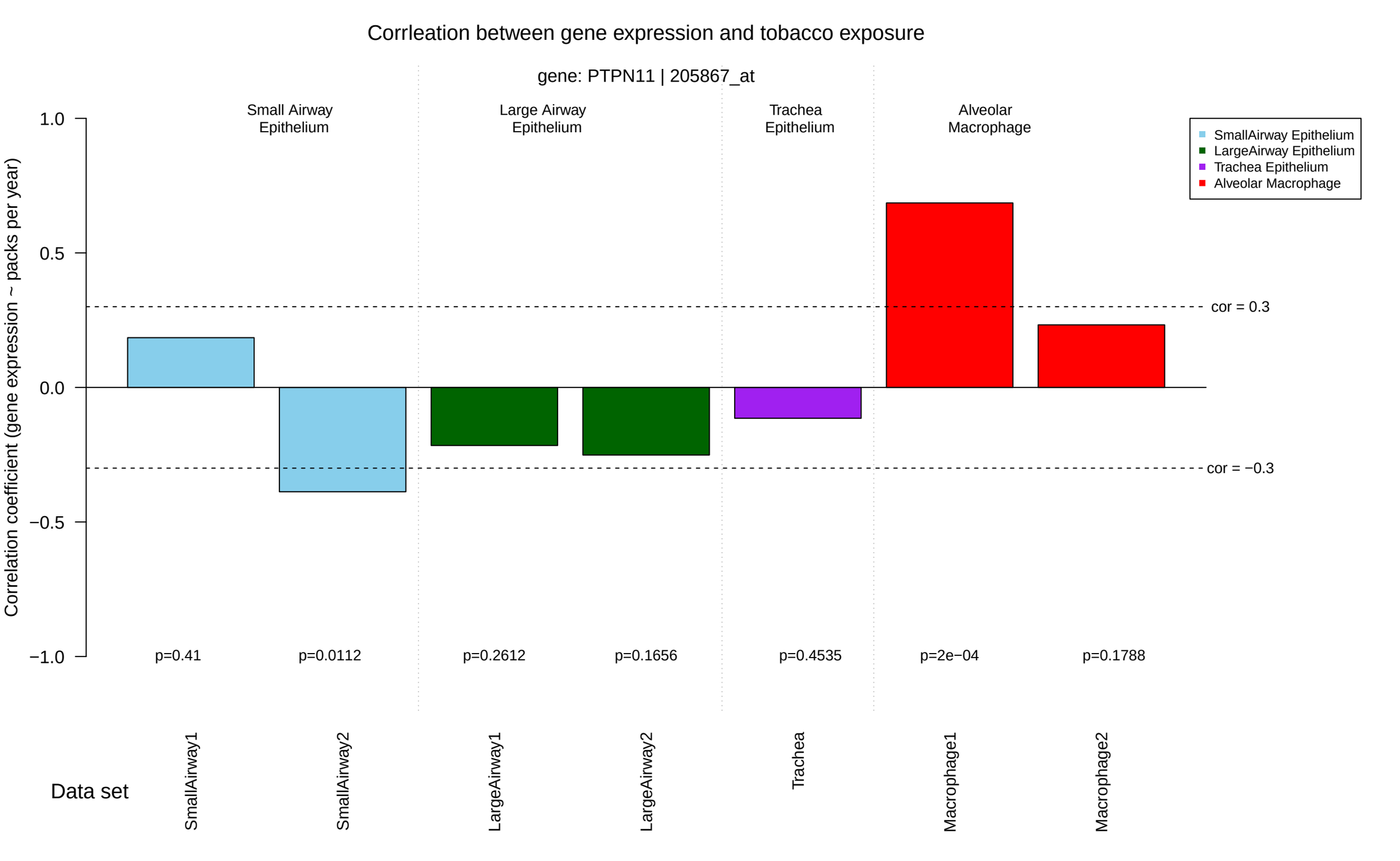

set: LargeAirway1 (p=0.193)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0221)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0356)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0336)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0135)
PTPN11 | 205867_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN11 | 205867_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.41
p=0.0112
p=0.2612
p=0.1656
p=0.4535
p=2e−04
p=0.1788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2